

Year 7 · Chapter 4 · Fractions and percentages: answers (teacher copy)

4.01 Fractions (booklet p. 3)

WE DO examples (teacher)

- Ex 1a: $\frac{7}{4}$
- Ex 1b: $1\frac{3}{4}$
- Ex 2a: $\frac{5}{15}$
- Ex 2b: $\frac{12}{16}$
- Ex 3a: $\frac{5}{6}$
- Ex 3b: $\frac{3}{2} = 1\frac{1}{2}$

Easy · Set A (quick drill)

- | | | |
|-------------------|-------------------|-------------------|
| 1. $\frac{7}{4}$ | 5. $\frac{9}{2}$ | 9. $\frac{3}{4}$ |
| 2. $\frac{22}{9}$ | 6. $\frac{17}{6}$ | 10. $\frac{1}{3}$ |
| 3. $\frac{28}{5}$ | 7. $\frac{3}{4}$ | 11. $\frac{2}{3}$ |
| 4. $\frac{22}{7}$ | 8. $\frac{2}{3}$ | 12. $\frac{2}{5}$ |

Easy · Set B

- $1\frac{3}{4}$
- $8\frac{2}{3}$
- $3\frac{2}{7}$
- $5\frac{1}{2}$
- $3\frac{2}{5}$
- $1\frac{1}{2}$

Medium · Set C

- 5
- 16
- 16
- 36
- 25
- 15

Challenging · Set D

- $\frac{5}{6}$
- $\frac{3}{4}$
- $\frac{3}{4}$
- $\frac{4}{3}$
- $\frac{1}{2}$
- $\frac{4}{7}$

Skill drill 1 (Easy basics)

- | | | |
|--------------------|---------------------|--------------------|
| 1. $\frac{1}{2}$ | 8. $\frac{3}{10}$ | 15. $\frac{1}{10}$ |
| 2. $\frac{11}{12}$ | 9. $\frac{5}{8}$ | 16. $\frac{5}{6}$ |
| 3. $\frac{1}{3}$ | 10. $\frac{1}{3}$ | 17. $\frac{1}{4}$ |
| 4. $\frac{1}{3}$ | 11. $\frac{11}{12}$ | 18. $\frac{3}{5}$ |
| 5. $\frac{1}{2}$ | 12. $\frac{9}{10}$ | 19. $\frac{3}{4}$ |
| 6. $\frac{1}{2}$ | 13. $\frac{2}{3}$ | 20. $\frac{5}{8}$ |
| 7. $\frac{1}{12}$ | 14. $\frac{3}{5}$ | 21. $\frac{3}{8}$ |

- $\frac{3}{4}$
- $\frac{2}{3}$
- $\frac{1}{8}$
- $\frac{17}{3}$
- $\frac{21}{10}$
- $3\frac{1}{2}$
- $\frac{37}{6}$

- $\frac{32}{5}$
- $2\frac{2}{3}$
- $\frac{57}{10}$
- $4\frac{5}{12}$
- $\frac{17}{4}$
- $\frac{5}{3}$
- $\frac{19}{10}$

- $6\frac{1}{3}$
- $\frac{18}{5}$
- $3\frac{2}{5}$
- $5\frac{1}{2}$
- $1\frac{1}{2}$

Skill drill 2 (Medium basics)

- 48
- 28
- 20
- 9
- 4
- 40
- 12
- 20
- 2
- 8
- 24
- 60
- 5
- 3
- 24

- 6
- 30
- 72
- 50
- 18
- $\frac{1}{2}$
- $\frac{2}{3}$
- $\frac{8}{9}$
- $\frac{2}{3}$
- $\frac{3}{10}$
- $\frac{1}{15}$

- $\frac{3}{5}$
- $\frac{2}{9}$
- $\frac{1}{4}$
- $\frac{1}{9}$
- $\frac{1}{3}$
- $\frac{1}{6}$
- $\frac{9}{11}$
- $\frac{4}{5}$
- $\frac{1}{2}$
- $\frac{5}{8}$

Extension

- $\frac{15}{30}$
- $\frac{10}{35}$

Exit ticket

- Level 1: $\frac{17}{5}$
- Level 2: 10
- Level 3: $\frac{3}{4}$

4.02 Ordering fractions (booklet p. 9)

WE DO examples (teacher)

- Ex 1a: $\frac{5}{8}$
- Ex 1b: $\frac{2}{3}$
- Ex 2a: $\frac{3}{10}, \frac{2}{5}, \frac{7}{10}, \frac{4}{5}$
- Ex 2b: $\frac{1}{3}, \frac{1}{2}, \frac{7}{12}, \frac{11}{12}$
- Ex 3a: $\frac{7}{12}, \frac{2}{3}, \frac{3}{4}, \frac{5}{6}$
- Ex 3b: $\frac{5}{4}, 1\frac{1}{3}, \frac{11}{8}$

Easy · Set A (quick drill)

- | | | |
|-------------------|------------------|--------------------|
| 1. $\frac{5}{7}$ | 5. $\frac{5}{6}$ | 9. $\frac{3}{4}$ |
| 2. $\frac{1}{3}$ | 6. $\frac{3}{4}$ | 10. $\frac{5}{6}$ |
| 3. $\frac{9}{10}$ | 7. $\frac{3}{5}$ | 11. $\frac{3}{10}$ |
| 4. $\frac{1}{2}$ | 8. $\frac{1}{2}$ | 12. $\frac{4}{7}$ |

Easy · Set B

- $\frac{6}{12}$

2. $\frac{8}{12}$
3. $\frac{9}{12}$
4. $\frac{4}{10}$
5. $\frac{10}{12}$
6. $\frac{9}{24}$

Medium · Set C

1. $\frac{3}{10}, \frac{2}{5}, \frac{7}{10}, \frac{4}{5}$
2. $\frac{1}{3}, \frac{1}{2}, \frac{7}{12}, \frac{11}{12}$
3. $\frac{3}{8}, \frac{1}{2}, \frac{5}{8}, \frac{3}{4}$
4. $\frac{4}{9}, \frac{1}{2}, \frac{2}{3}, \frac{5}{6}$

Challenging · Set D

1. $\frac{7}{12}, \frac{2}{3}, \frac{3}{4}, \frac{5}{6}$
2. $\frac{4}{15}, \frac{1}{2}, \frac{3}{5}, \frac{7}{10}$
3. $1\frac{1}{6}, \frac{5}{4}, 1\frac{1}{3}, \frac{11}{8}$
4. e.g. $\frac{5}{12}$

Skill drill 1 (Easy basics)

- | | |
|--------------------|---------------------------------------|
| 1. $\frac{3}{4}$ | 12. $\frac{3}{7}$ |
| 2. $\frac{7}{10}$ | 13. $\frac{9}{10}$ |
| 3. $\frac{1}{5}$ | 14. $\frac{2}{4}$ |
| 4. $\frac{6}{10}$ | 15. $\frac{1}{3}$ |
| 5. $\frac{5}{6}$ | 16. $\frac{2}{7}$ |
| 6. $\frac{9}{10}$ | 17. $\frac{5}{8}$ |
| 7. $\frac{11}{12}$ | 18. $\frac{3}{7}$ |
| 8. $\frac{4}{9}$ | 19. $\frac{4}{8}$ |
| 9. $\frac{4}{10}$ | 20. $\frac{4}{6}$ |
| 10. $\frac{1}{2}$ | 21. $\frac{4}{8}$ and $\frac{7}{8}$ |
| 11. $\frac{4}{5}$ | 22. $\frac{8}{12}$ and $\frac{6}{12}$ |

Skill drill 2 (Medium basics)

- | | | |
|--------------------|---------------------|--|
| 1. $\frac{9}{12}$ | 11. $\frac{10}{12}$ | 21. $\frac{2}{4}, \frac{8}{12}, \frac{3}{4}$ |
| 2. $\frac{7}{8}$ | 12. $\frac{1}{2}$ | 22. $\frac{1}{4}, \frac{4}{12}, \frac{10}{12}$ |
| 3. $\frac{3}{4}$ | 13. $\frac{5}{12}$ | 23. $\frac{1}{8}, \frac{2}{5}, \frac{3}{6}$ |
| 4. $\frac{2}{3}$ | 14. $\frac{4}{5}$ | 24. $\frac{1}{3}, \frac{4}{10}, \frac{1}{2}$ |
| 5. $\frac{1}{2}$ | 15. $\frac{2}{5}$ | 25. $\frac{1}{3}, \frac{4}{10}, \frac{6}{8}$ |
| 6. $\frac{8}{10}$ | 16. $\frac{5}{6}$ | 26. $\frac{5}{12}, \frac{1}{2}, \frac{10}{12}$ |
| 7. $\frac{3}{4}$ | 17. $\frac{2}{6}$ | 27. $\frac{1}{3}, \frac{3}{8}, \frac{2}{5}$ |
| 8. $\frac{4}{5}$ | 18. $\frac{6}{10}$ | 28. $\frac{3}{5}, \frac{3}{4}, \frac{4}{5}$ |
| 9. $\frac{7}{10}$ | 19. $\frac{8}{12}$ | 29. $\frac{1}{2}, \frac{2}{3}, \frac{5}{6}$ |
| 10. $\frac{9}{10}$ | 20. $\frac{7}{12}$ | 30. $\frac{1}{4}, \frac{2}{4}, \frac{5}{6}$ |

Extension

1. $\frac{3}{7}, \frac{4}{9}, \frac{1}{2}$
2. $\frac{7}{24}$

Exit ticket

1. Level 1: $\frac{7}{9}$
2. Level 2: $\frac{4}{5}$
3. Level 3: $\frac{2}{3}, \frac{3}{4}, \frac{5}{6}$

4.03 Adding and subtracting fractions

(booklet p. 15)

WE DO examples (teacher)

1. Ex 1a: 1
2. Ex 1b: $\frac{3}{7}$
3. Ex 2a: $\frac{7}{10}$
4. Ex 2b: $\frac{11}{20}$
5. Ex 3a: $\frac{5}{6}$
6. Ex 3b: $1\frac{7}{12}$

Easy · Set A (quick drill)

- | | | |
|------------------|-------------------|--------------------|
| 1. 1 | 5. $1\frac{2}{5}$ | 9. $\frac{3}{4}$ |
| 2. $\frac{7}{9}$ | 6. $\frac{1}{2}$ | 10. $\frac{1}{2}$ |
| 3. $\frac{2}{5}$ | 7. $\frac{3}{5}$ | 11. $\frac{7}{12}$ |
| 4. $\frac{1}{2}$ | 8. $\frac{1}{3}$ | 12. $\frac{1}{12}$ |

Easy · Set B

1. $\frac{3}{4}$
2. $\frac{1}{2}$
3. $\frac{7}{10}$
4. $\frac{1}{6}$
5. $\frac{1}{8}$
6. $\frac{7}{8}$

Medium · Set C

1. $\frac{5}{6}$
2. $\frac{11}{20}$
3. $\frac{11}{12}$
4. $\frac{3}{10}$
5. $1\frac{7}{12}$
6. $\frac{25}{72}$

Challenging · Set D

1. $1\frac{19}{30}$
2. $1\frac{1}{12}$
3. $\frac{3}{8}$
4. $\frac{4}{15}$

Skill drill 1 (Easy basics)

- | | | |
|-------------------|-------------------|--------------------|
| 1. 1 | 6. $1\frac{2}{5}$ | 10. $\frac{2}{11}$ |
| 2. $1\frac{1}{3}$ | 7. $\frac{8}{11}$ | 11. $\frac{1}{12}$ |
| 3. 1 | 8. $\frac{2}{5}$ | 12. $\frac{1}{4}$ |
| 4. 1 | 9. $1\frac{4}{7}$ | 13. $\frac{5}{12}$ |

$$\begin{array}{l} 14. 1\frac{7}{10} \\ 15. 1\frac{4}{7} \\ 16. \frac{3}{4} \\ 17. 1\frac{1}{4} \\ 18. 1\frac{2}{9} \\ 19. \frac{3}{5} \\ 20. 1\frac{1}{3} \\ 21. \frac{2}{3} \end{array}$$

$$\begin{array}{l} 22. \frac{3}{11} \\ 23. \frac{3}{7} \\ 24. \frac{1}{4} \\ 25. 1\frac{1}{3} \\ 26. \frac{2}{3} \\ 27. 1 \\ 28. \frac{1}{4} \\ 29. \frac{5}{6} \\ 30. \frac{7}{10} \end{array}$$

$$\begin{array}{l} 31. 1\frac{1}{15} \\ 32. \frac{8}{15} \\ 33. \frac{7}{12} \\ 34. 1\frac{1}{4} \\ 35. \frac{4}{5} \\ 36. 1 \\ 37. \frac{1}{8} \\ 38. 1 \\ 39. 1 \\ 40. \frac{9}{10} \end{array}$$

$$\begin{array}{l} 1. 9\frac{2}{3} \\ 2. 4\frac{1}{4} \\ 3. 4\frac{1}{2} \\ 4. 2\frac{3}{5} \\ 5. 4 \\ 6. 1\frac{3}{4} \end{array}$$

Medium · Set C

$$\begin{array}{l} 1. 8\frac{7}{10} \\ 2. 1\frac{1}{2} \\ 3. 5\frac{1}{4} \\ 4. 2\frac{5}{12} \\ 5. 4\frac{5}{12} \\ 6. 2\frac{7}{12} \end{array}$$

Challenging · Set D

$$\begin{array}{l} 1. 1\frac{7}{12} \\ 2. 1\frac{7}{10} \\ 3. 6\frac{7}{12} \\ 4. 1\frac{3}{4} \text{ m} \end{array}$$

Skill drill 1 (Easy basics)

$$\begin{array}{l} 1. 2\frac{1}{2} \\ 2. 13 \\ 3. 9 \\ 4. 5\frac{3}{4} \\ 5. 5\frac{1}{5} \\ 6. 2 \\ 7. 6\frac{1}{5} \\ 8. 6\frac{1}{3} \\ 9. 12\frac{2}{3} \\ 10. 5 \\ 11. 11\frac{1}{10} \\ 12. 6\frac{1}{2} \\ 13. 5 \end{array}$$

$$\begin{array}{l} 14. 2\frac{1}{3} \\ 15. 14 \\ 16. 3 \\ 17. 5\frac{1}{6} \\ 18. 1 \\ 19. 2\frac{1}{5} \\ 20. 6\frac{1}{7} \\ 21. 7\frac{8}{9} \\ 22. 4\frac{1}{6} \\ 23. 3\frac{1}{2} \\ 24. 8\frac{3}{4} \\ 25. 4\frac{1}{5} \end{array}$$

$$\begin{array}{l} 26. 7\frac{3}{8} \\ 27. 1\frac{1}{10} \\ 28. 4\frac{1}{3} \\ 29. 6\frac{1}{7} \\ 30. 6\frac{2}{5} \\ 31. 6\frac{1}{3} \\ 32. 6\frac{3}{4} \\ 33. 2\frac{1}{3} \\ 34. 6\frac{4}{9} \\ 35. 1\frac{1}{2} \\ 36. 2\frac{4}{5} \end{array}$$

Skill drill 2 (Medium basics)

$$\begin{array}{l} 1. 5\frac{9}{10} \\ 2. 11\frac{1}{4} \\ 3. 8\frac{7}{10} \\ 4. 4\frac{5}{8} \\ 5. 7\frac{3}{4} \\ 6. 7\frac{17}{30} \\ 7. 7\frac{11}{24} \\ 8. 6\frac{14}{15} \end{array}$$

$$\begin{array}{l} 9. 8\frac{11}{12} \\ 10. 7\frac{23}{24} \\ 11. 7\frac{5}{8} \\ 12. 8\frac{1}{4} \\ 13. 1\frac{1}{2} \\ 14. 6\frac{4}{5} \\ 15. 2\frac{1}{2} \\ 16. 4\frac{2}{5} \end{array}$$

$$\begin{array}{l} 17. 4\frac{2}{3} \\ 18. 3\frac{2}{3} \\ 19. 2\frac{3}{5} \\ 20. \frac{1}{2} \\ 21. 1\frac{4}{5} \\ 22. 7\frac{2}{3} \\ 23. 1\frac{1}{2} \\ 24. 2\frac{2}{5} \end{array}$$

Extension

$$\begin{array}{l} 1. 1\frac{1}{12} \\ 2. 3\frac{1}{2} \end{array}$$

Skill drill 2 (Medium basics)

$$\begin{array}{l} 1. 1\frac{1}{30} \\ 2. \frac{5}{6} \\ 3. \frac{5}{6} \\ 4. \frac{3}{8} \\ 5. 1\frac{1}{12} \\ 6. \frac{1}{4} \\ 7. 1\frac{2}{5} \\ 8. 1\frac{7}{15} \\ 9. 1\frac{1}{6} \\ 10. 1\frac{13}{30} \end{array}$$

$$\begin{array}{l} 11. \frac{3}{10} \\ 12. \frac{7}{12} \\ 13. \frac{7}{12} \\ 14. \frac{5}{8} \\ 15. \frac{21}{40} \\ 16. \frac{1}{10} \\ 17. \frac{17}{24} \\ 18. \frac{1}{3} \\ 19. \frac{1}{2} \\ 20. \frac{3}{4} \end{array}$$

$$\begin{array}{l} 21. \frac{1}{2} \\ 22. \frac{2}{3} \\ 23. \frac{5}{12} \\ 24. \frac{5}{8} \\ 25. \frac{11}{24} \\ 26. \frac{7}{12} \\ 27. \frac{3}{8} \\ 28. \frac{5}{12} \end{array}$$

Extension

$$\begin{array}{l} 1. \frac{11}{15} \\ 2. \frac{1}{3} + \frac{1}{4} \end{array}$$

Exit ticket

$$\begin{array}{l} 1. \text{Level 1: } \frac{5}{7} \\ 2. \text{Level 2: } \frac{1}{3} \\ 3. \text{Level 3: } 1\frac{4}{15} \end{array}$$

4.04 Adding and subtracting mixed numerals (booklet p. 21)

WE DO examples (teacher)

$$\begin{array}{l} 1. \text{Ex 1a: } 8\frac{7}{10} \\ 2. \text{Ex 1b: } 5\frac{1}{4} \\ 3. \text{Ex 2a: } 9\frac{2}{3} \\ 4. \text{Ex 2b: } 1\frac{1}{2} \\ 5. \text{Ex 3a: } 1\frac{7}{12} \\ 6. \text{Ex 3b: } 2\frac{13}{20} \end{array}$$

Easy · Set A (quick drill)

$$\begin{array}{l} 1. 3\frac{3}{5} \\ 2. 5\frac{1}{2} \\ 3. 3\frac{3}{7} \\ 4. 8\frac{1}{2} \end{array}$$

$$\begin{array}{l} 5. 4\frac{1}{3} \\ 6. 3 \\ 7. 3\frac{2}{3} \\ 8. 2\frac{2}{5} \end{array}$$

$$\begin{array}{l} 9. 2\frac{3}{4} \\ 10. 3 \\ 11. 3\frac{1}{4} \\ 12. 3\frac{1}{3} \end{array}$$

Easy · Set B

Exit ticket

1. Level 1: $6\frac{2}{3}$
2. Level 2: $5\frac{3}{5}$
3. Level 3: $1\frac{7}{12}$

4.05 Fraction of a quantity (booklet p. 27)

WE DO examples (teacher)

1. Ex 1a: 7
2. Ex 1b: 9
3. Ex 2a: 16 h
4. Ex 2b: 1600 m
5. Ex 3a: 135 min
6. Ex 3b: 50 s

Easy · Set A (quick drill)

- | | | |
|--------|-------|--------|
| 1. 7 | 5. 20 | 9. 15 |
| 2. 3.5 | 6. 9 | 10. 8 |
| 3. 8 | 7. 6 | 11. 20 |
| 4. 12 | 8. 8 | 12. 15 |

Easy · Set B

1. 24
2. 35
3. 105
4. 750
5. 12
6. 40

Medium · Set C

1. 16 h
2. 1600 m
3. 135 min
4. 105 mm
5. 50 s
6. 125 cm

Challenging · Set D

1. 12 boys
2. \$240
3. 36
4. $\frac{3}{4}$ h by 3 min

Skill drill 1 (Easy basics)

- | | | |
|--------|--------|--------|
| 1. 6 | 15. 2 | 29. 21 |
| 2. 3 | 16. 4 | 30. 16 |
| 3. 6 | 17. 11 | 31. 36 |
| 4. 12 | 18. 2 | 32. 32 |
| 5. 2 | 19. 2 | 33. 30 |
| 6. 3 | 20. 3 | 34. 70 |
| 7. 3 | 21. 3 | 35. 33 |
| 8. 8 | 22. 2 | 36. 18 |
| 9. 3 | 23. 10 | 37. 16 |
| 10. 7 | 24. 7 | 38. 44 |
| 11. 8 | 25. 50 | 39. 40 |
| 12. 11 | 26. 14 | 40. 21 |
| 13. 9 | 27. 72 | |
| 14. 9 | 28. 15 | |

Skill drill 2 (Medium basics)

- | | | |
|-----------|-----------|-------------------|
| 1. 760 g | 7. 375 m | 13. 10 c |
| 2. 500 g | 8. 30 min | 14. 16 c |
| 3. 18 h | 9. 16 h | 15. 920 g |
| 4. 50 g | 10. 12 h | 16. $\frac{4}{5}$ |
| 5. 36 min | 11. 250 m | 17. $\frac{1}{5}$ |
| 6. 20 cm | 12. 250 g | |

18. $\frac{5}{12}$

19. $\frac{2}{3}$

20. $\frac{1}{6}$

21. $\frac{1}{5}$

22. $\frac{8}{15}$

23. $\frac{13}{24}$

24. $\frac{2}{5}$

25. $\frac{9}{10}$

26. $\frac{5}{12}$

27. $\frac{1}{5}$

28. $\frac{11}{24}$

29. $\frac{2}{3}$

30. $\frac{4}{5}$

31. $\frac{4}{15}$

Extension

1. 33.75
2. 160 L

Exit ticket

1. Level 1: 9
2. Level 2: 60
3. Level 3: 90 min

4.06 Multiplying fractions (booklet p. 33)

WE DO examples (teacher)

1. Ex 1a: $\frac{2}{15}$
2. Ex 1b: 6
3. Ex 2a: $\frac{5}{14}$
4. Ex 2b: $\frac{1}{40}$
5. Ex 3a: $\frac{3}{4}$
6. Ex 3b: $10\frac{8}{9}$

Easy · Set A (quick drill)

- | | | |
|-------------------|-------------------|-------------------|
| 1. $\frac{2}{15}$ | 5. 6 | 9. $1\frac{1}{2}$ |
| 2. $\frac{1}{21}$ | 6. 10 | 10. $\frac{1}{4}$ |
| 3. $\frac{3}{8}$ | 7. $\frac{1}{6}$ | 11. 4 |
| 4. $\frac{4}{15}$ | 8. $\frac{3}{10}$ | 12. $\frac{1}{6}$ |

Easy · Set B

1. $\frac{5}{14}$
2. $\frac{1}{6}$
3. $\frac{1}{2}$
4. $\frac{1}{40}$
5. $\frac{1}{6}$
6. $\frac{3}{5}$

Medium · Set C

1. $\frac{3}{4}$
2. $1\frac{1}{10}$
3. $2\frac{1}{4}$
4. $1\frac{1}{3}$
5. $1\frac{3}{4}$
6. $1\frac{7}{20}$

Challenging · Set D

1. $10\frac{8}{9}$
2. $2\frac{1}{3}$
3. $1\frac{1}{6}$

4. $1\frac{5}{6}$ cups

Skill drill 1 (Easy basics)

- | | | |
|--------------------|--------------------|--------------------|
| 1. $\frac{5}{36}$ | 14. $\frac{5}{36}$ | 27. $2\frac{2}{5}$ |
| 2. $\frac{1}{9}$ | 15. $\frac{2}{9}$ | 28. $1\frac{3}{5}$ |
| 3. $\frac{6}{35}$ | 16. $\frac{1}{24}$ | 29. $7\frac{1}{5}$ |
| 4. $\frac{7}{16}$ | 17. $\frac{4}{15}$ | 30. $7\frac{6}{7}$ |
| 5. $\frac{4}{25}$ | 18. $\frac{2}{9}$ | 31. 6 |
| 6. $\frac{1}{18}$ | 19. $\frac{3}{16}$ | 32. $3\frac{1}{3}$ |
| 7. $\frac{1}{4}$ | 20. $\frac{1}{5}$ | 33. 1 |
| 8. $\frac{5}{56}$ | 21. $\frac{6}{35}$ | 34. $5\frac{1}{2}$ |
| 9. $\frac{1}{4}$ | 22. $\frac{1}{27}$ | 35. $2\frac{3}{4}$ |
| 10. $\frac{1}{10}$ | 23. $\frac{1}{12}$ | 36. $1\frac{1}{5}$ |
| 11. $\frac{2}{5}$ | 24. $\frac{5}{28}$ | 37. $2\frac{1}{2}$ |
| 12. $\frac{3}{10}$ | 25. 1 | 38. $1\frac{3}{7}$ |
| 13. $\frac{2}{5}$ | 26. 2 | 39. 4 |
| | | 40. $3\frac{1}{9}$ |

Skill drill 2 (Medium basics)

- | | | |
|--------------------|---------------------|---------------------|
| 1. $\frac{3}{20}$ | 10. $\frac{1}{14}$ | 19. $3\frac{8}{9}$ |
| 2. $\frac{5}{28}$ | 11. $\frac{4}{49}$ | 20. $3\frac{9}{10}$ |
| 3. $\frac{3}{14}$ | 12. $\frac{1}{6}$ | 21. 4 |
| 4. $\frac{16}{35}$ | 13. $\frac{1}{3}$ | 22. $2\frac{7}{9}$ |
| 5. $\frac{2}{21}$ | 14. $\frac{1}{5}$ | 23. $11\frac{1}{9}$ |
| 6. $\frac{3}{5}$ | 15. $\frac{4}{35}$ | 24. $12\frac{5}{6}$ |
| 7. $\frac{1}{15}$ | 16. $\frac{2}{15}$ | 25. $8\frac{1}{6}$ |
| 8. $\frac{1}{6}$ | 17. $12\frac{5}{6}$ | 26. $3\frac{3}{4}$ |
| 9. $\frac{5}{28}$ | 18. $5\frac{1}{15}$ | 27. $5\frac{1}{2}$ |
| | | 28. $9\frac{7}{9}$ |

Extension

- $\frac{1}{10}$
- $9\frac{1}{3}$ m²

Exit ticket

- Level 1: $\frac{3}{20}$
- Level 2: $\frac{1}{6}$
- Level 3: $3\frac{1}{3}$

4.07 Dividing fractions (booklet p. 39)

WE DO examples (teacher)

- Ex 1a: $\frac{4}{3}$, $\frac{1}{5}$, $\frac{2}{5}$
- Ex 1b: 50
- Ex 2a: $\frac{1}{6}$
- Ex 2b: 2
- Ex 3a: 2
- Ex 3b: $1\frac{1}{2}$

Easy · Set A (quick drill)

- | | | |
|------------------|------------------|------------------|
| 1. $\frac{4}{3}$ | 2. $\frac{5}{2}$ | 3. $\frac{1}{7}$ |
|------------------|------------------|------------------|

- | | | |
|------------------|------------------|--------------------|
| 4. 9 | 7. 12 | 10. $\frac{1}{2}$ |
| 5. $\frac{3}{8}$ | 8. $\frac{1}{6}$ | 11. $2\frac{1}{2}$ |
| 6. $\frac{2}{3}$ | 9. 3 | 12. 6 |

Easy · Set B

- 50
- $\frac{1}{6}$
- $1\frac{1}{2}$
- 9
- $\frac{2}{5}$
- 12

Medium · Set C

- 2
- 6
- $\frac{32}{35}$
- $1\frac{5}{12}$
- $\frac{4}{5}$
- $\frac{5}{6}$

Challenging · Set D

- 3
- 2
- $1\frac{1}{2}$
- 10 pieces

Skill drill 1 (Easy basics)

- | | | |
|-------------------|---------------------|--------|
| 1. $\frac{2}{10}$ | 13. $\frac{7}{5}$ | 25. 12 |
| 2. $\frac{1}{5}$ | 14. $\frac{2}{8}$ | 26. 14 |
| 3. $\frac{3}{4}$ | 15. $\frac{6}{4}$ | 27. 12 |
| 4. $\frac{10}{5}$ | 16. $\frac{12}{3}$ | 28. 36 |
| 5. $\frac{7}{4}$ | 17. 3 | 29. 40 |
| 6. $\frac{1}{4}$ | 18. $\frac{1}{12}$ | 30. 15 |
| 7. $\frac{1}{8}$ | 19. $\frac{2}{3}$ | 31. 42 |
| 8. $\frac{4}{10}$ | 20. $\frac{7}{10}$ | 32. 49 |
| 9. 6 | 21. $\frac{12}{11}$ | 33. 20 |
| 10. $\frac{8}{7}$ | 22. $\frac{1}{9}$ | 34. 12 |
| 11. $\frac{1}{3}$ | 23. $\frac{4}{8}$ | 35. 25 |
| 12. $\frac{1}{6}$ | 24. $\frac{11}{7}$ | 36. 18 |
| | | 37. 24 |
| | | 38. 16 |
| | | 39. 21 |
| | | 40. 48 |

Skill drill 2 (Medium basics)

- | | | |
|--------------------|----------------------|----------------------|
| 1. 2 | 11. $1\frac{1}{8}$ | 20. $1\frac{1}{2}$ |
| 2. $\frac{2}{3}$ | 12. $2\frac{1}{4}$ | 21. $1\frac{1}{25}$ |
| 3. $\frac{7}{18}$ | 13. $\frac{1}{2}$ | 22. $2\frac{2}{15}$ |
| 4. $\frac{1}{2}$ | 14. $\frac{5}{21}$ | 23. $1\frac{2}{5}$ |
| 5. $2\frac{1}{2}$ | 15. $\frac{20}{21}$ | 24. $3\frac{8}{9}$ |
| 6. $3\frac{1}{3}$ | 16. $\frac{9}{14}$ | 25. $2\frac{2}{25}$ |
| 7. $\frac{1}{2}$ | 17. $1\frac{2}{5}$ | 26. $2\frac{7}{10}$ |
| 8. 1 | 18. $1\frac{11}{14}$ | 27. 1 |
| 9. $2\frac{1}{2}$ | 19. 1 | 28. $1\frac{34}{35}$ |
| 10. $2\frac{2}{3}$ | | |

Extension

1. $\frac{2}{3}$
2. 5

Exit ticket

1. Level 1: $\frac{8}{5}$
2. Level 2: 4
3. Level 3: 2

4.08 Fraction problems (booklet p. 45)

WE DO examples (teacher)

1. Ex 1a: $6\frac{7}{12}$ L
2. Ex 1b: $1\frac{1}{4}$ L
3. Ex 2a: $\frac{1}{12}$
4. Ex 2b: 1 goat
5. Ex 3a: $\frac{4}{15}$
6. Ex 3b: 64 pages

Easy · Set A

1. $\frac{5}{8}$
2. $\frac{1}{2}$
3. 24
4. $\frac{1}{3}$
5. $\frac{5}{8}$
6. 12

Easy · Set A2 (quick drill)

1. $\frac{3}{4}$
2. $\frac{3}{5}$
3. $\frac{1}{3}$
4. $\frac{1}{4}$
5. $\frac{5}{8}$
6. $\frac{1}{2}$

Easy · Set B

1. $2\frac{5}{6}$ m
2. $4\frac{1}{2}$ kg
3. $\frac{5}{12}$

Medium · Set C

1. $4\frac{7}{12}$ L
2. 1
3. 24 L
4. \$30
5. 12
6. 480 km

Challenging · Set D

1. 100
2. $4\frac{9}{10}$ m
3. 63
4. No: they are equal (40)

Skill drill 1 (Easy basics)

1. $\frac{2}{5}$
2. $\frac{1}{4}$
3. $\frac{1}{3}$

4. $\frac{1}{6}$
5. $\frac{4}{5}$
6. $\frac{2}{3}$
7. $\frac{1}{3}$
8. $\frac{2}{5}$
9. $\frac{3}{5}$
10. $\frac{1}{3}$
11. $\frac{3}{4}$
12. $\frac{1}{3}$
13. $\frac{2}{3}$
14. $\frac{1}{5}$
15. $\frac{1}{3}$
16. $\frac{11}{12}$
17. $\frac{3}{8}$
18. $\frac{2}{3}$
19. 6 kilograms

Skill drill 2 (Medium basics)

1. \$6
2. \$15
3. \$16
4. \$57
5. \$33
6. \$4
7. \$27
8. \$12
9. \$40
10. \$28
11. 18
12. 70
13. 6
14. 10
15. 84
16. 45

Extension

1. 24
2. 300 mL

Exit ticket

1. Level 1: $\frac{11}{15}$
2. Level 2: 9
3. Level 3: \$70

4.09 Fractions on a calculator (booklet p. 53)

WE DO examples (teacher)

1. Ex 1a: $\frac{11}{12}$
2. Ex 1b: $1\frac{2}{5}$
3. Ex 2a: $3\frac{2}{5}$; 3.4
4. Ex 2b: $\frac{25}{72}$
5. Ex 3a: $12\frac{1}{4}$
6. Ex 3b: $5\frac{1}{16}$

Easy · Set A (quick drill)

1. $\frac{11}{12}$
2. $1\frac{19}{30}$
3. $\frac{19}{36}$
4. $\frac{25}{72}$
5. $4\frac{7}{12}$
6. $2\frac{5}{6}$
7. $\frac{29}{36}$
8. $\frac{13}{30}$
9. $\frac{3}{8}$
10. $1\frac{1}{2}$

Easy · Set B

1. $\frac{1}{40}$
2. $1\frac{5}{12}$
3. $10\frac{8}{9}$

4. $2\frac{7}{24}$
5. 225
6. 28

Medium · Set C

1. $3\frac{2}{5}$
2. $5\frac{5}{8}$
3. $\frac{19}{8}$
4. 0.35
5. $\frac{5}{8}$
6. 0.3125

Challenging · Set D

1. $12\frac{1}{4}$
2. $5\frac{9}{16}$
3. $2\frac{4}{15}$
4. $\approx 4 \times 2 = 8$; $8\frac{11}{80}$

Skill drill 1 (Easy basics)

- | | | |
|----------------------|-----------------------|-----------------------|
| 1. $\frac{3}{22}$ | 13. $\frac{19}{52}$ | 25. $1\frac{5}{22}$ |
| 2. $\frac{17}{24}$ | 14. 1 | 26. $\frac{9}{112}$ |
| 3. $\frac{5}{8}$ | 15. $1\frac{1}{14}$ | 27. $1\frac{1}{8}$ |
| 4. $1\frac{71}{234}$ | 16. $\frac{13}{45}$ | 28. $\frac{2}{3}$ |
| 5. $\frac{35}{247}$ | 17. $1\frac{46}{117}$ | 29. $1\frac{81}{280}$ |
| 6. $\frac{23}{40}$ | 18. $\frac{9}{35}$ | 30. $\frac{54}{187}$ |
| 7. $\frac{13}{15}$ | 19. $\frac{29}{45}$ | 31. $\frac{7}{39}$ |
| 8. $\frac{53}{84}$ | 20. $\frac{205}{238}$ | 32. $\frac{2}{75}$ |
| 9. $\frac{181}{240}$ | 21. $\frac{4}{35}$ | 33. $\frac{6}{35}$ |
| 10. $\frac{77}{190}$ | 22. $\frac{121}{126}$ | 34. $\frac{12}{91}$ |
| 11. $1\frac{11}{21}$ | 23. $\frac{8}{35}$ | 35. $\frac{3}{16}$ |
| 12. $\frac{37}{171}$ | 24. $\frac{270}{323}$ | 36. $\frac{16}{45}$ |

Skill drill 2 (Medium basics)

- | | | |
|----------------------|----------------------|----------------------|
| 1. $\frac{2}{3}$ | 11. $\frac{23}{56}$ | 21. $1\frac{13}{14}$ |
| 2. $\frac{5}{12}$ | 12. $\frac{49}{54}$ | 22. $1\frac{5}{11}$ |
| 3. $1\frac{13}{72}$ | 13. $8\frac{8}{11}$ | 23. $2\frac{11}{15}$ |
| 4. $\frac{55}{98}$ | 14. $3\frac{2}{3}$ | 24. $1\frac{5}{8}$ |
| 5. $\frac{11}{49}$ | 15. $1\frac{5}{13}$ | 25. $2\frac{1}{6}$ |
| 6. $\frac{57}{280}$ | 16. $6\frac{7}{12}$ | 26. $5\frac{7}{10}$ |
| 7. $\frac{8}{21}$ | 17. $1\frac{13}{15}$ | 27. $7\frac{2}{3}$ |
| 8. $\frac{1}{3}$ | 18. $8\frac{4}{15}$ | 28. $5\frac{5}{9}$ |
| 9. $\frac{5}{8}$ | 19. $4\frac{5}{9}$ | |
| 10. $\frac{37}{144}$ | 20. $2\frac{8}{11}$ | |

Extension

1. $\frac{31}{32}$: it gets closer to 1
2. $\frac{13}{15}$

Exit ticket

1. Level 1: $\frac{31}{35}$
2. Level 2: $3\frac{1}{6}$
3. Level 3: 4

4.10 Percentages, fractions and decimals

(booklet p. 59)

WE DO examples (teacher)

1. Ex 1a: $\frac{13}{100}$; 0.13
2. Ex 1b: $\frac{2}{25}$; 0.08
3. Ex 2a: $\frac{3}{40}$; 0.075
4. Ex 2b: 70%
5. Ex 3a: 92%
6. Ex 3b: 4.5%

Easy · Set A (quick drill)

- | | | |
|---------------------|-------------------|---------|
| 1. $\frac{13}{100}$ | 5. $\frac{7}{20}$ | 9. 20% |
| 2. $\frac{2}{25}$ | 6. $1\frac{1}{5}$ | 10. 0.4 |
| 3. $\frac{1}{2}$ | 7. $\frac{1}{4}$ | 11. 75% |
| 4. $\frac{3}{4}$ | 8. 60% | 12. 7% |

Easy · Set B

1. 0.13
2. 0.08
3. 0.45
4. 1.5
5. 0.025
6. 0.075

Medium · Set C

1. 70%
2. 92%
3. 75%
4. 60%
5. 4.5%
6. 130%

Challenging · Set D

1. 0.375, 37.5%
2. $33\frac{1}{3}\%$
3. $\frac{1}{8}$
4. $\frac{3}{8}$, 38%, $0.4 = \frac{2}{5}$

Skill drill 1 (Easy basics)

- | | | |
|--------------------|--------------------|----------|
| 1. $\frac{1}{4}$ | 11. $\frac{9}{10}$ | 21. 0.99 |
| 2. $\frac{3}{4}$ | 12. $\frac{3}{5}$ | 22. 1.11 |
| 3. $\frac{7}{20}$ | 13. $\frac{1}{5}$ | 23. 0.34 |
| 4. $\frac{2}{25}$ | 14. $\frac{1}{10}$ | 24. 0.22 |
| 5. $\frac{9}{20}$ | 15. $\frac{1}{25}$ | 25. 0.62 |
| 6. $\frac{1}{100}$ | 16. $\frac{7}{10}$ | 26. 0.68 |
| 7. $\frac{4}{5}$ | 17. $\frac{1}{50}$ | 27. 1.1 |
| 8. $\frac{3}{20}$ | 18. $\frac{1}{20}$ | 28. 0.1 |
| 9. $\frac{2}{5}$ | 19. $\frac{3}{10}$ | 29. 0.14 |
| 10. $\frac{3}{25}$ | 20. $\frac{1}{2}$ | 30. 0.47 |
| | | 31. 0.35 |
| | | 32. 0.59 |
| | | 33. 0.76 |
| | | 34. 1.17 |
| | | 35. 0.46 |
| | | 36. 1.18 |

Skill drill 2 (Medium basics)

- | | | |
|---------|-------------------------|-------------------------|
| 1. 60% | 15. 76% | 24. 5%, $\frac{1}{20}$ |
| 2. 32% | 16. 42% | 25. 2%, $\frac{1}{50}$ |
| 3. 70% | 17. 30%, $\frac{3}{10}$ | 26. 10%, $\frac{1}{10}$ |
| 4. 34% | 18. 40%, $\frac{2}{5}$ | 27. 4%, $\frac{1}{25}$ |
| 5. 80% | 19. 20%, $\frac{1}{5}$ | 28. 80%, $\frac{4}{5}$ |
| 6. 36% | 20. 12%, $\frac{3}{25}$ | 29. 70%, $\frac{7}{10}$ |
| 7. 50% | 21. 50%, $\frac{1}{2}$ | 30. 90%, $\frac{9}{10}$ |
| 8. 10% | 22. 75%, $\frac{3}{4}$ | 31. 45%, $\frac{9}{20}$ |
| 9. 25% | 23. 25%, $\frac{1}{4}$ | |
| 10. 40% | | |
| 11. 65% | | |
| 12. 92% | | |
| 13. 16% | | |
| 14. 26% | | |

Extension

- $66\frac{2}{3}\%$
- $\frac{37}{300}$

Exit ticket

- Level 1: $\frac{1}{4}$
- Level 2: 7%
- Level 3: 18.75%

4.11 Percentage of a quantity (booklet p. 65)

WE DO examples (teacher)

- Ex 1a: 35
- Ex 1b: 21
- Ex 2a: \$9
- Ex 2b: 84
- Ex 3a: 60 kg
- Ex 3b: \$175

Easy · Set A (quick drill)

- | | | |
|-------|-------|--------|
| 1. 35 | 5. 9 | 9. 30 |
| 2. 43 | 6. 3 | 10. 3 |
| 3. 21 | 7. 7 | 11. 11 |
| 4. 9 | 8. 32 | 12. 30 |

Easy · Set B

- \$9
- 84
- 75
- 27
- 9
- 60

Medium · Set C

- 60 kg
- \$63
- 200 m
- 81 min
- 51
- \$10

Challenging · Set D

- \$175
- 9
- \$46
- Equal: both 21

Skill drill 1 (Easy basics)

- | | | |
|-------|-------|--------|
| 1. 28 | 5. 25 | 9. 2 |
| 2. 52 | 6. 11 | 10. 36 |
| 3. 84 | 7. 29 | 11. 14 |
| 4. 17 | 8. 14 | 12. 6 |

- | | | |
|--------|---------|---------|
| 13. 26 | 23. 45 | 33. 45 |
| 14. 4 | 24. 74 | 34. 156 |
| 15. 59 | 25. 24 | 35. 180 |
| 16. 46 | 26. 72 | 36. 17 |
| 17. 34 | 27. 78 | 37. 64 |
| 18. 23 | 28. 20 | 38. 8 |
| 19. 9 | 29. 112 | 39. 435 |
| 20. 17 | 30. 14 | 40. 13 |
| 21. 35 | 31. 29 | |
| 22. 82 | 32. 26 | |

Skill drill 2 (Medium basics)

- | | | |
|------------|------------|-------------|
| 1. \$72 | 11. \$82.5 | 21. \$20 |
| 2. \$36 | 12. \$35 | 22. \$112 |
| 3. \$212.5 | 13. \$15 | 23. \$108 |
| 4. \$66 | 14. \$259 | 24. \$136 |
| 5. \$186 | 15. \$315 | 25. \$229.5 |
| 6. \$88 | 16. \$52.5 | 26. \$45 |
| 7. \$95 | 17. \$97.5 | 27. \$162 |
| 8. \$72.5 | 18. \$198 | 28. \$204 |
| 9. \$49.5 | 19. \$234 | |
| 10. \$23 | 20. \$54 | |

Extension

- \$57.60
- No: 20% then 10% is 28% off

Exit ticket

- Level 1: 8
- Level 2: 70
- Level 3: \$51

4.12 Expressing as a fraction or percentage (booklet p. 71)

WE DO examples (teacher)

- Ex 1a: $\frac{17}{20}$; 85%
- Ex 1b: 75%
- Ex 2a: $\frac{1}{5}$
- Ex 2b: 20%
- Ex 3a: 37.5%
- Ex 3b: Mia 85%, Jake 84%: Mia

Easy · Set A (quick drill)

- | | | |
|--------------------|------------------|---------|
| 1. $\frac{3}{5}$ | 4. $\frac{2}{3}$ | 8. 45% |
| 2. $\frac{3}{4}$ | 5. $\frac{2}{5}$ | 9. 48% |
| 3. $\frac{17}{20}$ | 6. $\frac{1}{5}$ | 10. 34% |
| | 7. 30% | 11. 75% |
| | | 12. 80% |

Easy · Set B

- 70%
- 75%
- 85%
- 65%
- 90%
- 37.5%

Medium · Set C

- $\frac{1}{5}$, 20%
- $\frac{3}{8}$, 37.5%
- $\frac{1}{8}$, 12.5%
- $\frac{2}{25}$, 8%
- $\frac{3}{10}$, 30%

6. $\frac{1}{4}$, 25%

Challenging · Set D

- 1. Mia (85% vs 84%)
- 2. 25%
- 3. 25%
- 4. 57.1%

Skill drill 1 (Easy basics)

- | | |
|---------|--------------------|
| 1. 10% | 15. 55% |
| 2. 50% | 16. 56% |
| 3. 75% | 17. 42% |
| 4. 50% | 18. 25% |
| 5. 25% | 19. 20% |
| 6. 20% | 20. 88% |
| 7. 40% | 21. 35% |
| 8. 90% | 22. 70% |
| 9. 92% | 23. 80% |
| 10. 36% | 24. 10% |
| 11. 60% | 25. $\frac{3}{10}$ |
| 12. 32% | 26. $\frac{5}{16}$ |
| 13. 20% | 27. $\frac{1}{2}$ |
| 14. 30% | |

- 28. $\frac{3}{20}$
- 29. $\frac{17}{30}$
- 30. $\frac{11}{24}$
- 31. $\frac{29}{30}$
- 32. $\frac{11}{20}$
- 33. $\frac{13}{24}$
- 34. $\frac{11}{20}$
- 35. $\frac{11}{15}$
- 36. $\frac{11}{12}$

37. $\frac{1}{16}$
38. $\frac{13}{16}$

39. $\frac{9}{20}$
40. $\frac{1}{3}$

Skill drill 2 (Medium basics)

- | | | |
|-----------|-----------|------------|
| 1. 5% | 11. 6.25% | 21. 60% |
| 2. 25% | 12. 25% | 22. 65% |
| 3. 10% | 13. 80% | 23. 64% |
| 4. 3.33% | 14. 80% | 24. 81.25% |
| 5. 2.5% | 15. 75% | 25. 96% |
| 6. 15% | 16. 42.5% | 26. 97.5% |
| 7. 20% | 17. 97.5% | 27. 40% |
| 8. 50% | 18. 35% | 28. 72.5% |
| 9. 10% | 19. 62% | |
| 10. 37.5% | 20. 85% | |

Extension

- 1. 24%
- 2. 20%

Exit ticket

- 1. Level 1: $\frac{3}{4}$
- 2. Level 2: 84%
- 3. Level 3: 20%